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LAB EXPERIMENT: TIME SERIES ANALYSIS AND DECOMPOSITION

AIM

To perform time series analysis and decomposition on the monthly average temperature of a city using Python.

ALGORITHM

1. Start the program.
2. Import Pandas, Matplotlib, and seasonal_decompose from statsmodels.
3. Create monthly temperature data for two complete yearly cycles.
4. Generate a monthly date index using Pandas.
5. Create a time-series object using the temperature values and date index.
6. Display the monthly temperature data.
7. Plot the original monthly average temperature time series.
8. Apply additive seasonal decomposition with period = 12.
9. Display the Observed, Trend, Seasonal, and Residual components.
10. Interpret the plotted components.
11. Stop the program.

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PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

temperature = [
    25, 27, 29, 32, 34, 33,
    32, 31, 30, 29, 27, 25,
    25, 27, 29, 32, 34, 33,
    32, 31, 30, 29, 27, 25
]

dates = pd.date_range(
    start="2025-01-01",
    periods=24,
    freq="ME"
)

data = pd.Series(
    temperature,
    index=dates
)

print("Monthly Temperature:")
print(data)

plt.figure(figsize=(10, 5))
plt.plot(data.index, data.values, marker="o")
plt.title("Monthly Average Temperature")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.grid()
plt.show()

result = seasonal_decompose(
    data,
    model="additive",
    period=12
)

result.plot()
plt.show()
```

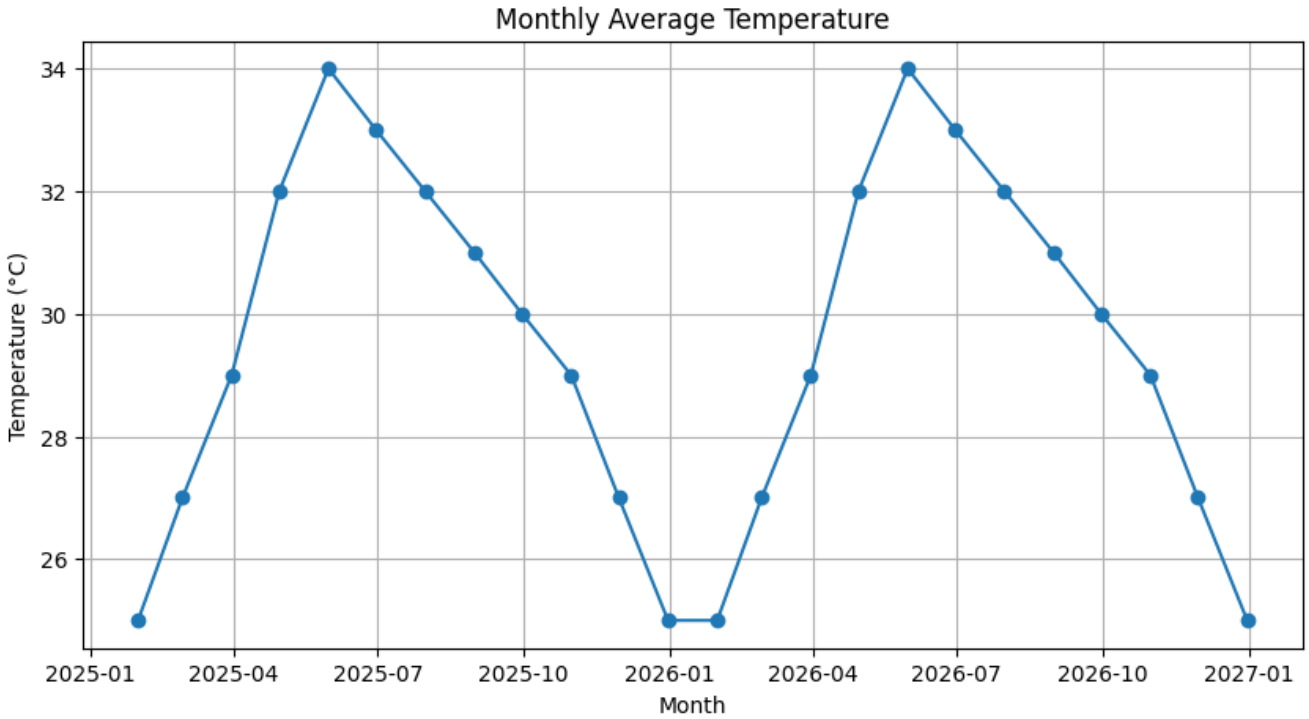
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OUTPUT

Monthly Temperature Data

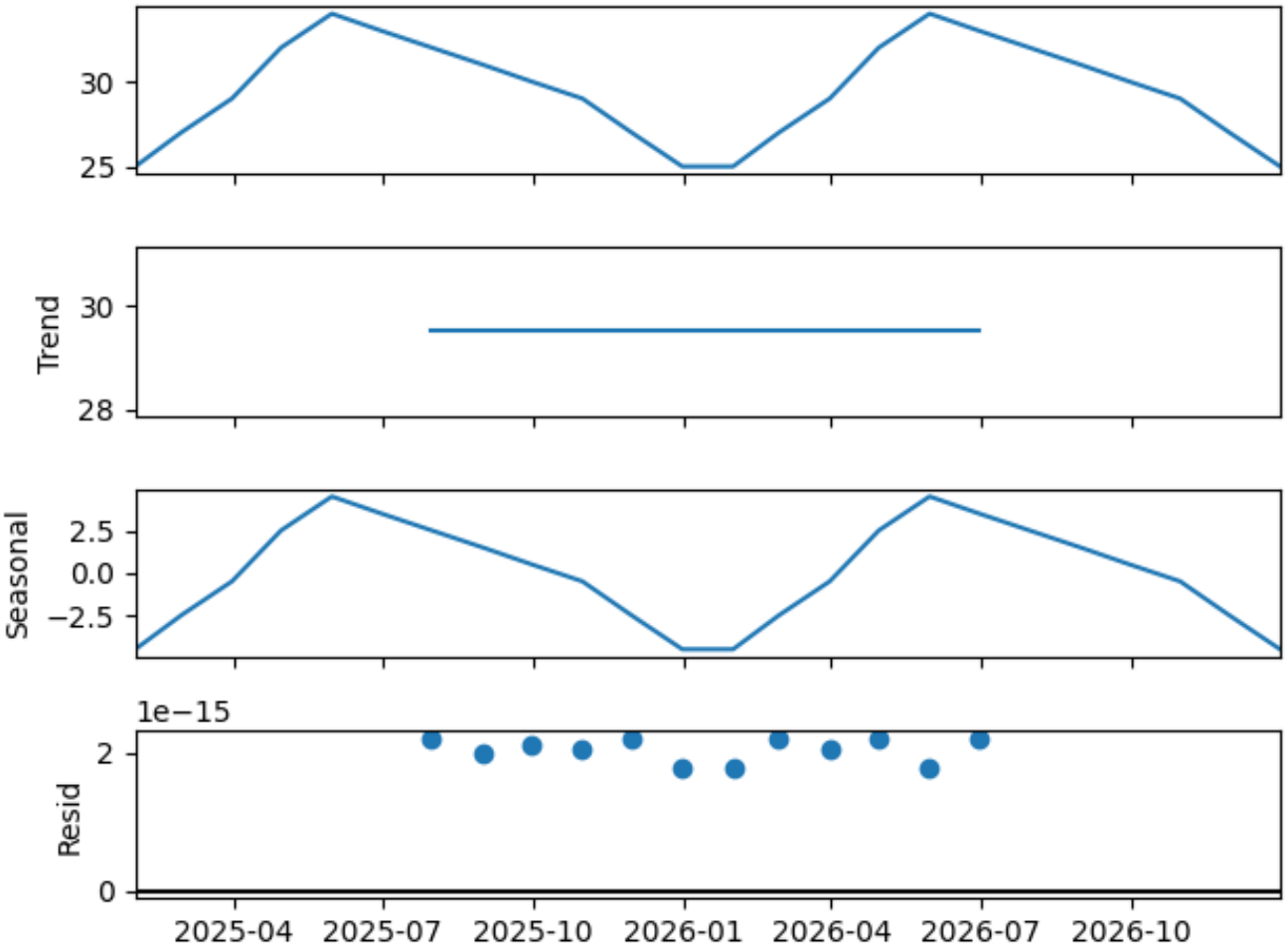
```
Monthly Temperature:
2025-01-31    25
2025-02-28    27
2025-03-31    29
2025-04-30    32
2025-05-31    34
2025-06-30    33
2025-07-31    32
2025-08-31    31
2025-09-30    30
2025-10-31    29
2025-11-30    27
2025-12-31    25
2026-01-31    25
2026-02-28    27
2026-03-31    29
2026-04-30    32
2026-05-31    34
2026-06-30    33
2026-07-31    32
2026-08-31    31
2026-09-30    30
2026-10-31    29
2026-11-30    27
2026-12-31    25
Freq: ME, dtype: int64
```

Original Time Series



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OUTPUT - TIME SERIES DECOMPOSITION



OBSERVATION

The monthly temperature shows a repeating seasonal pattern. The decomposition separates the original series into observed, trend, seasonal, and residual components. The residual values are very close to zero because the two yearly cycles follow the same repeated temperature pattern.

RESULT

Thus, the monthly temperature time series was successfully plotted and decomposed into observed, trend, seasonal, and residual components using Python.